Linear Circuits 2
AC Load Measurement Lab Report

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Objectives

For this project, the objective was to determine the impedance and power factor of a hidden AC load at a specific frequency of the V peak to peak, as the current of the load using a resistor due to the accessibility of the black box being only wires. Then, we were tasked to find the phase angle and determine if it was inductive or capacitive.

We were given the following circuit (abstractly):

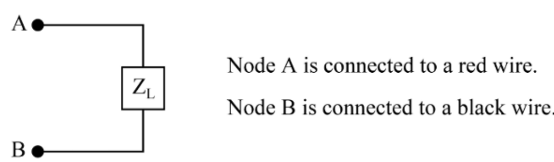

Figure 1: Abstract representation of the hidden AC load Z_L. Node A is connected to a red wire; Node B is connected to a black wire.

Experimental Plan

To start off, we found the current of the whole series circuit. This involved putting our multimeter in series with node B and ground while setting the multimeter to current mode. We pursued finding the voltages of each element and to do that we utilized a test resistor. After this, we then proceeded to use Ohm's Law to find the impedance of both the resistor and the black box. We then need to find the phase difference to find out if the box is inductive or capacitive in nature, which will tell us what components are inside the box. To find the phase angle, we used the oscilloscope to find it with the reference being the original input AC voltage.

Experimental Results

Now for the experimental results, we measured the voltage across the shunt resistor:

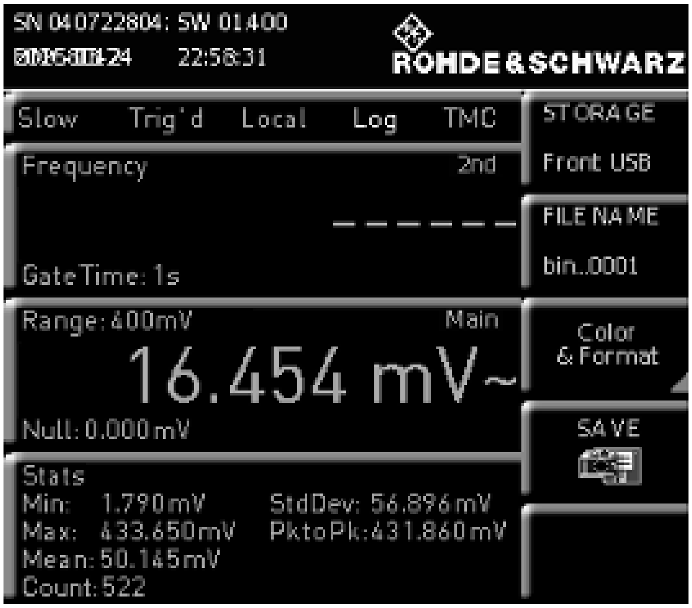

Figure 2: Measured voltage across the shunt resistor, 16.454 mV

For the current, we measured the current leaving the black box and entering the same shunt resistor:

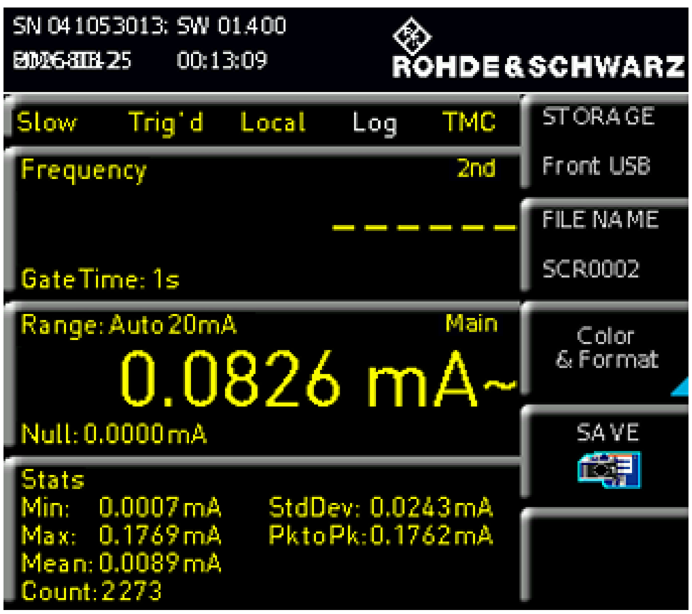

Figure 3: Measured current leaving the black box, 0.0826 mA

Using these measurements, some hand calculations were made:

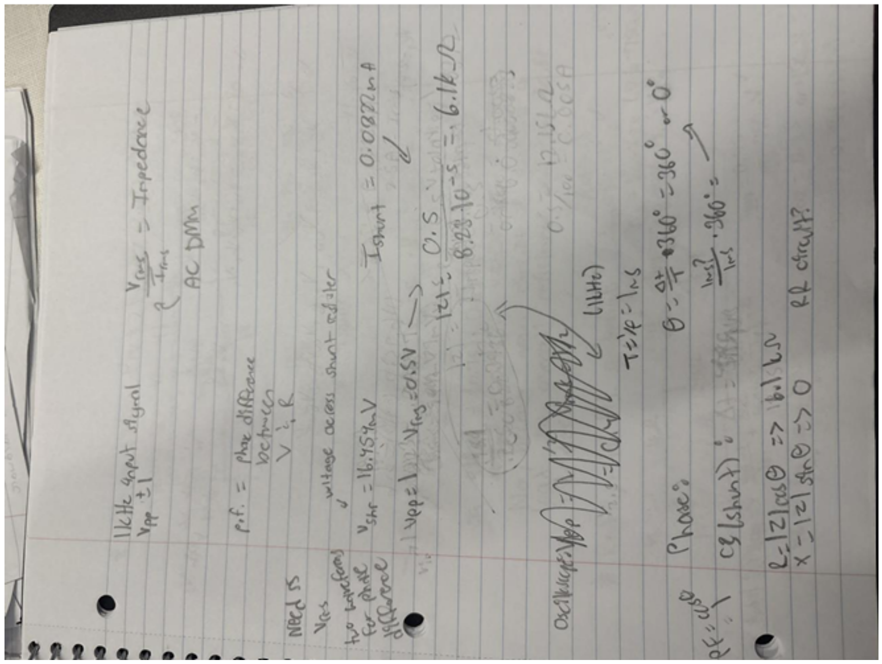

Figure 4: Hand calculations for impedance, phase difference, and power factor

$$Z = 6.1\,\mathrm{k}\Omega + j0$$

Results Comparison

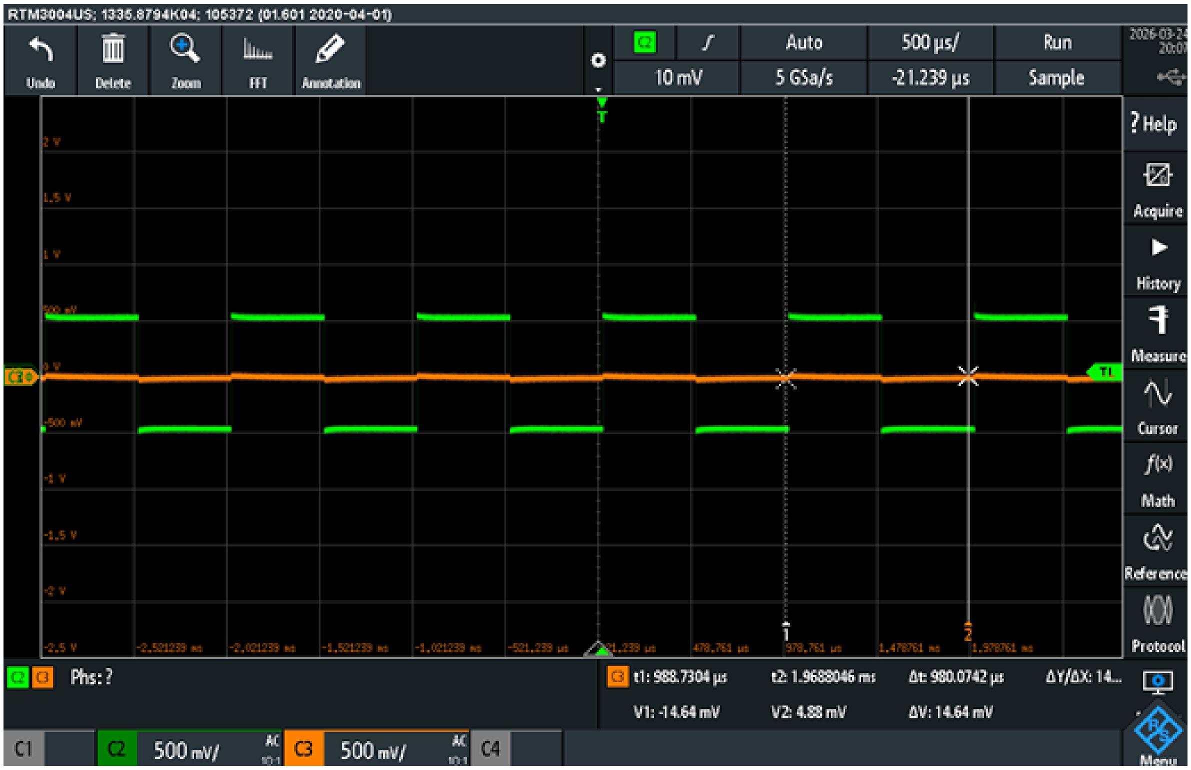

Figure 5: Voltage (C2) and current (C3) waveform

For this section, after finding out that our phase angle is zero, we determined that our black box is not capacitive or inductive in nature. Thus, this means our mystery circuit is purely resistive, aka an RR circuit. The power factor is 1, which means physically, the voltage and current are perfectly in phase, which can be seen in the image above.

Conclusion

In this experiment, the objective was to identify the characteristics of the unknown AC load by determining its impedance, power factor, and its nature. This is done by measuring the voltage and current across the concealed circuit and a test resistor, allowing us to analyze the circuit's behavior. Our measurements revealed an impedance of 6.1 kΩ and a phase angle of zero, which led to the conclusion of the concealed circuit being purely resistive.